

DigitalOcean Deployment Guide — Nucleus AI

Deploy Nucleus AI on DigitalOcean using **App Platform** (PaaS) or **Droplets** (VPS).

Prerequisites

- DigitalOcean account
- `doctl` CLI installed and authenticated
- Docker installed locally (for Droplet option)
- Domain name pointed to DigitalOcean

Option A: App Platform (Easiest)

1. Create Container Registry

```
doctl registry create nucleus-ai-registry --region nyc3
doctl registry login
```

2. Push Images

```
# Build and tag
docker build -f docker/production/Dockerfile.backend -t registry.digitalocean.com/nucleus-ai-registry/backend:latest .
docker build -f docker/production/Dockerfile.frontend \
  --build-arg NEXT_PUBLIC_API_URL=https://api.yourdomain.com \
  -t registry.digitalocean.com/nucleus-ai-registry/frontend:latest .

# Push
docker push registry.digitalocean.com/nucleus-ai-registry/backend:latest
docker push registry.digitalocean.com/nucleus-ai-registry/frontend:latest
```

3. Create Managed PostgreSQL

```
doctl databases create nucleus-db \
  --engine pg \
  --version 16 \
  --size db-s-2vcpu-4gb \
  --region nyc3 \
  --num-nodes 1
```

4. Create App via Spec

Create `do-app-spec.yaml` :

```

name: nucleus-ai
region: nyc
services:
- name: backend
  image:
    registry_type: DOCR
    repository: backend
    tag: latest
  instance_count: 2
  instance_size_slug: professional-xs
  http_port: 8000
  health_check:
    http_path: /health
  envs:
    - key: APP_ENV
      value: production
    - key: OPENAI_API_KEY
      type: SECRET
      value: sk-your-key
    - key: DATABASE_URL
      value: ${nucleus-db.DATABASE_URL}
  routes:
    - path: /api
    - path: /health

- name: frontend
  image:
    registry_type: DOCR
    repository: frontend
    tag: latest
  instance_count: 1
  instance_size_slug: professional-xs
  http_port: 3000
  health_check:
    http_path: /api/health
  envs:
    - key: NODE_ENV
      value: production
    - key: NEXTAUTH_SECRET
      type: SECRET
      value: your-secret
  routes:
    - path: /

databases:
- name: nucleus-db
  engine: PG
  version: "16"
  production: true
  cluster_name: nucleus-db

```

```
doctl apps create --spec do-app-spec.yaml
```

5. Custom Domain

```

doctl apps update <APP_ID> --spec do-app-spec.yaml
# Add domain via console: Apps → Settings → Domains

```

DigitalOcean App Platform provides free SSL certificates automatically.

Option B: Droplet (VPS)

1. Create Droplet

```
doctl compute droplet create nucleus-ai \  
  --image ubuntu-24-04-x64 \  
  --size s-4vcpu-8gb \  
  --region nyc3 \  
  --ssh-keys <YOUR_SSH_KEY_ID>
```

2. Server Setup

```
ssh root@<DROPLET_IP>  
  
# Install Docker  
curl -fsSL https://get.docker.com | sh  
  
# Install Docker Compose  
apt install docker-compose-plugin -y  
  
# Clone and configure  
git clone https://github.com/your-org/nucleus-ai.git /opt/nucleus-ai  
cd /opt/nucleus-ai  
cp .env.production.example .env.production  
nano .env.production  
  
# Set up SSL  
apt install certbot -y  
certbot certonly --standalone -d yourdomain.com  
cp /etc/letsencrypt/live/yourdomain.com/fullchain.pem nginx/ssl/  
cp /etc/letsencrypt/live/yourdomain.com/privkey.pem nginx/ssl/  
  
# Deploy  
docker compose -f docker-compose.prod.yml --env-file .env.production up -d
```

3. Enable Firewall

```
ufw allow 22/tcp  
ufw allow 80/tcp  
ufw allow 443/tcp  
ufw enable
```

Cost Estimation (Monthly)

Option	Service	Estimated Cost
App Platform	2× pro-xs + 1× pro-xs	~\$36
	Managed PostgreSQL	~\$30
	Container Registry	~\$5
Droplet	s-4vcpu-8gb	~\$48
Shared	Qdrant Cloud Starter	~\$25
Total (App Platform)		~\$96/mo
Total (Droplet)		~\$73/mo